

```
6. # Access the third item (index 2)
7. middle_price = prices[2] # 144.22
8.
9. # Access the last item
10. closing_price = prices[-1] # 145.03
11.
12. # Access the second-to-last item
13. previous_price = prices[-2] # 143.86
14.
```

Python's zero-based indexing (starting from 0 instead of 1) can be confusing at first, but it follows a logical pattern: the index represents the offset from the beginning of the list.

An easy way to understand this is to think of the index as "how many steps do I need to move from the start to reach the element?" For the first element, you don't need to move at all (0 steps), while for the second element, you need to move 1 step.

The negative indices are particularly useful in trading applications, where the most recent data (like the latest price) is often at the end of the list.

## Slicing Lists: Extracting Time Periods

Sometimes you need a subset of your data, like the last 20 days of prices for a moving average calculation. Python's slicing syntax makes this easy:

```
1. prices = [142.56, 143.78, 144.22, 143.86, 145.03, 146.89, 148.56, 147.27]
2.
3. # Get prices from index 2 to index 5 (exclusive)
4. mid_week_prices = prices[2:5] # [144.22, 143.86, 145.03]
5.
6. # Get the first 3 prices
7. early_prices = prices[:3] # [142.56, 143.78, 144.22]
8.
9. # Get the last 3 prices
10. recent_prices = prices[-3:] # [146.89, 148.56, 147.27]
11.
12. # Get every other price (step of 2)
```